Program 1: Connect to a Database, Store and Retrieve Information

File name: DatabaseDemo.java | **Type:** Java + JDBC

SQL setup

```
CREATE DATABASE labdb;
USE labdb;
CREATE TABLE student (
    id INT PRIMARY KEY,
    name VARCHAR(50)
);
```

Java code

```
import java.sql.*;

public class DatabaseDemo {
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");

        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/labdb",
            "root", "your_password"
        );

        Statement st = con.createStatement();
        st.executeUpdate(
            "INSERT INTO student VALUES (1, 'Arun')"
        );

        ResultSet rs = st.executeQuery("SELECT * FROM student");
        while (rs.next()) {
            System.out.println(rs.getInt("id") + " "
                    + rs.getString("name"));
        }

        con.close();
    }
}
```

Note: Add MySQL Connector/J to the project build path. Change the password and avoid running the same INSERT repeatedly because id 1 is a primary key.

Program 2: Find the Factorial Using JSP

File name: factorial.jsp **|** **Type:** JSP

Enter a number from 0 to 20. The page calculates its factorial and displays the result.

```
<html>
<body>
<h2>Factorial Program</h2>

<form method="post">
    Enter a number:
    <input type="number" name="num" min="0" max="20" required>
    <input type="submit" value="Find Factorial">
</form>

<%
String s = request.getParameter("num");

if (s != null) {
    try {
        int n = Integer.parseInt(s);

        if (n < 0 || n > 20) {
            out.println("Enter a number from 0 to 20");
        } else {
            long fact = 1;
            for (int i = 1; i <= n; i++) {
                fact = fact * i;
            }
            out.println("Factorial = " + fact);
        }
    } catch (NumberFormatException e) {
        out.println("Invalid number");
    }
}
%>
</body>
</html>
```

Sample output: input 5 → Factorial = 120.

Program 3: Registration Form with Database Storage and Retrieval

File name: register.jsp, save.jsp, view.jsp | **Type:** JSP + JDBC

SQL setup

```
USE labdb;
CREATE TABLE users (
    name VARCHAR(50),
    password VARCHAR(100),
    email VARCHAR(100),
    country VARCHAR(50)
);
```

File 1: register.jsp

```
<html>
<body>
<h2>Registration Form</h2>
<form action="save.jsp" method="post">
    Name: <input name="name" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    Country: <input name="country" required><br><br>
    <input type="submit" value="Register">
</form>
</body>
</html>
```

File 2: save.jsp

```
<%@ page import="java.sql.*" %>
<html><body>
<%
String name = request.getParameter("name");
String password = request.getParameter("password");
String email = request.getParameter("email");
String country = request.getParameter("country");

try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/labdb",
        "root", "your_password"
    );

    PreparedStatement ps = con.prepareStatement(
        "INSERT INTO users VALUES (?, ?, ?, ?)"
    );
    ps.setString(1, name);
    ps.setString(2, password);
    ps.setString(3, email);
    ps.setString(4, country);
    ps.executeUpdate();

    out.println("Registration successful!");
    ps.close();
    con.close();
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
<br><a href="view.jsp">View Users</a>
</body></html>
```

File 3: view.jsp

```
<%@ page import="java.sql.*" %>
<html><body>
<h2>Registered Users</h2>
<table border="1">
<tr><th>Name</th><th>Email</th><th>Country</th></tr>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/labdb",
        "root", "your_password"
    );

    ResultSet rs = con.createStatement().executeQuery(
        "SELECT name, email, country FROM users"
    );
    while (rs.next()) {
%>
<tr>
<td><%= rs.getString("name") %></td>
<td><%= rs.getString("email") %></td>
<td><%= rs.getString("country") %></td>
</tr>
<%
    }
    con.close();
} catch (Exception e) {
    out.println("Database error.");
}
%>
</table>
</body></html>
```

Note: This is a classroom demonstration only. Do not store real passwords as plain text; use secure password hashing in real applications.

Program 4: Use JavaBeans in JSP to Display Name and Age

File name: StudentBean.java, input.jsp, display.jsp | **Type:** JavaBean + JSP

File 1: StudentBean.java

```
package mybean;

public class StudentBean {
    private String name;
    private int age;

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public int getAge() { return age; }
    public void setAge(int age) { this.age = age; }
}
```

File 2: input.jsp

```
<html><body>
<form action="display.jsp" method="post">
    Name: <input name="name"><br><br>
    Age: <input type="number" name="age"><br><br>
    <input type="submit" value="Submit">
</form>
</body></html>
```

File 3: display.jsp

```
<jsp:useBean id="s" class="mybean.StudentBean" scope="request" />
<jsp:setProperty name="s" property="*" />
<html><body>
<h2>Student Details</h2>
Name: <jsp:getProperty name="s" property="name" /><br>
Age: <jsp:getProperty name="s" property="age" />
</body></html>
```

Note: Place StudentBean.java in the Java source folder under package mybean. The package name and folder must match.

Program 5: Calculator Using a Servlet

File name: calculator.html, CalculatorServlet.java | **Type:** HTML + Servlet

File 1: calculator.html

```
<html><body>
<h2>Calculator</h2>
<form action="calc" method="post">
    Number 1: <input type="number" step="any" name="a" required><br><br>
    Number 2: <input type="number" step="any" name="b" required><br><br>
    Operation:
    <select name="op">
        <option>+</option><option>-</option>
        <option>*</option><option>/</option>
    </select>
    <input type="submit" value="Calculate">
</form>
</body></html>
```

File 2: CalculatorServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/calc")
public class CalculatorServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req,
            HttpServletResponse res)
            throws ServletException, IOException {

        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        try {
            double a = Double.parseDouble(req.getParameter("a"));
            double b = Double.parseDouble(req.getParameter("b"));
            String op = req.getParameter("op");
            double result;

            switch (op) {
                case "+": result = a + b; break;
                case "-": result = a - b; break;
                case "*": result = a * b; break;
                case "/":
                    if (b == 0) {
                        out.println("Cannot divide by zero");
                        return;
                    }
                    result = a / b;
                    break;
                default:
                    out.println("Invalid operator");
                    return;
            }
            out.println("<h2>Result = " + result + "</h2>");
        } catch (NumberFormatException e) {
            out.println("Please enter valid numbers.");
        }
    }
}
```

Program 6: Send Name and Email from JSP to Servlet and Another JSP

File name: input.jsp, ProcessServlet.java, result.jsp | **Type:** JSP + Servlet

File 1: input.jsp

```
<html><body>
<form action="process" method="post">
    Name: <input name="name" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    <input type="submit" value="Submit">
</form>
</body></html>
```

File 2: ProcessServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/process")
public class ProcessServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req,
            HttpServletResponse res)
            throws ServletException, IOException {

        req.setAttribute("name", req.getParameter("name"));
        req.setAttribute("email", req.getParameter("email"));
        req.getRequestDispatcher("result.jsp").forward(req, res);
    }
}
```

File 3: result.jsp

```
<%@ page import="java.net.URLEncoder" %>
<html><body>
<h2>Submitted Details</h2>
Name: <%= org.apache.commons.text.StringEscapeUtils.escapeHtml4(
    String.valueOf(request.getAttribute("name"))) %><br>
Email: <%= org.apache.commons.text.StringEscapeUtils.escapeHtml4(
    String.valueOf(request.getAttribute("email"))) %>
</body></html>
```

Note: This result.jsp uses Apache Commons Text for HTML escaping. Add that library to the project. If you prefer no extra library, use a small HTML-escape helper. Always escape user-submitted text before displaying it.

Program 7: Student CRUD Operations Using JDBC and Servlet

File name: StudentServlet.java | **Type:** Servlet + JDBC

SQL setup

```
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/student")
public class StudentServlet extends HttpServlet {
    Connection getCon() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/labdb",
            "root", "your_password"
        );
    }

    protected void doPost(HttpServletRequest req,
            HttpServletResponse res) throws IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        String action = req.getParameter("action");

        try (Connection con = getCon()) {
            int id = Integer.parseInt(req.getParameter("id"));

            if ("delete".equals(action)) {
                PreparedStatement ps = con.prepareStatement(
                    "DELETE FROM students WHERE id=?");
                ps.setInt(1, id);
                ps.executeUpdate();
            } else {
                String name = req.getParameter("name");
                int age = Integer.parseInt(req.getParameter("age"));
                String sql = "insert".equals(action)
                    ? "INSERT INTO students VALUES(?,?,?)"
                    : "UPDATE students SET name=?, age=? WHERE id=?";
                PreparedStatement ps = con.prepareStatement(sql);

                if ("insert".equals(action)) {
                    ps.setInt(1, id);
                    ps.setString(2, name);
                    ps.setInt(3, age);
                } else {
                    ps.setString(1, name);
                    ps.setInt(2, age);
                    ps.setInt(3, id);
                }
                ps.executeUpdate();
            }
            out.println("Operation successful.<br><a href='student'>View Students</a>");
        } catch (Exception e) {
            out.println("Error: " + e.getMessage());
        }
    }

    protected void doGet(HttpServletRequest req,
            HttpServletResponse res) throws IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        out.println("<h2>Student CRUD</h2>");
        out.println("<form method='post' action='student'>");
        out.println("ID: <input name='id' required><br>");
```

Program 8: JSP Registration and Login Authentication

File name: register.jsp, saveuser.jsp, login.jsp, checklogin.jsp | **Type:** JSP + JDBC

SQL setup

```
USE labdb;
CREATE TABLE loginusers (
    username VARCHAR(50) PRIMARY KEY,
    password VARCHAR(100)
);
```

File 1: register.jsp

```
<html><body>
<h2>Register</h2>
<form action="saveuser.jsp" method="post">
    Username: <input name="username" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    <input type="submit" value="Register">
</form>
<a href="login.jsp">Login</a>
</body></html>
```

File 2: saveuser.jsp

```
<%@ page import="java.sql.*" %>
<html><body>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/labdb", "root", "your_password");
    PreparedStatement ps = con.prepareStatement(
        "INSERT INTO loginusers VALUES (?,?)");
    ps.setString(1, request.getParameter("username"));
    ps.setString(2, request.getParameter("password"));
    ps.executeUpdate();
    out.println("Registration successful");
    con.close();
} catch (Exception e) {
    out.println("Registration failed");
}
%>
<br><a href="login.jsp">Login</a>
</body></html>
```

File 3: login.jsp

```
<html><body>
<h2>Login</h2>
<form action="checklogin.jsp" method="post">
    Username: <input name="username" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    <input type="submit" value="Login">
</form>
</body></html>
```

File 4: checklogin.jsp

```
<%@ page import="java.sql.*" %>
<html><body>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/labdb", "root", "your_password");
    PreparedStatement ps = con.prepareStatement(
        "SELECT * FROM loginusers WHERE username=? AND password=?");
    ps.setString(1, request.getParameter("username"));
    ps.setString(2, request.getParameter("password"));
    ResultSet rs = ps.executeQuery();

    if (rs.next()) {
        out.println("Login successful");
    } else {
        out.println("Invalid username or password");
    }
    con.close();
} catch (Exception e) {
    out.println("Login error");
}
%>
</body></html>
```

Note: This is a simplified lab-only example. For real applications, hash passwords and never store or compare plain-text passwords.

Program 9: Employee Details Using Servlet and JDBC

File name: employee.html, EmployeeServlet.java | **Type:** HTML + Servlet + JDBC

SQL setup

```
USE labdb;
CREATE TABLE employee (
    empid INT PRIMARY KEY,
    empname VARCHAR(50),
    deptname VARCHAR(50),
    mobile VARCHAR(15),
    salary DOUBLE
);
```

File 1: employee.html

```
<html><body>
<h2>Employee Details</h2>
<form action="employee" method="post">
    Employee ID: <input type="number" name="empid" required><br><br>
    Employee Name: <input name="empname" required><br><br>
    Department: <input name="deptname" required><br><br>
    Mobile: <input name="mobile" required><br><br>
    Salary: <input type="number" step="any" name="salary" required><br><br>
    <input type="submit" value="Save Employee">
</form>
</body></html>
```

File 2: EmployeeServlet.java

```
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/employee")
public class EmployeeServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req,
            HttpServletResponse res)
            throws ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            try (Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/labdb",
                    "root", "your_password")) {
                PreparedStatement ps = con.prepareStatement(
                    "INSERT INTO employee VALUES (?,?,?,?,?)");
                ps.setInt(1, Integer.parseInt(req.getParameter("empid")));
                ps.setString(2, req.getParameter("empname"));
                ps.setString(3, req.getParameter("deptname"));
                ps.setString(4, req.getParameter("mobile"));
                ps.setDouble(5, Double.parseDouble(req.getParameter("salary")));
                ps.executeUpdate();
                out.println("Employee added successfully<br>");

                ResultSet rs = con.createStatement().executeQuery(
                    "SELECT * FROM employee");
                out.println("<table border='1'><tr><th>ID</th><th>Name</th>"
                    + "<th>Department</th><th>Mobile</th><th>Salary</th></tr>");
                while (rs.next()) {
                    out.println("<tr><td>" + rs.getInt(1)
                        + "</td><td>" + rs.getString(2)
                        + "</td><td>" + rs.getString(3)
                        + "</td><td>" + rs.getString(4)
                        + "</td><td>" + rs.getDouble(5) + "</td></tr>");
                }
                out.println("</table>");
```

Program 10: HTML Form Processed by Servlet Using POST

File name: login.html, UserServlet.java | **Type:** HTML + Servlet

File 1: login.html

```
<html><body>
<h2>User Form</h2>
<form action="user" method="post">
    Name: <input name="name" required><br><br>
    Password: <input type="password" name="password" required><br><br>
    <input type="submit" value="Submit">
</form>
</body></html>
```

File 2: UserServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/user")
public class UserServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req,
            HttpServletResponse res)
            throws ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        String name = req.getParameter("name");
        out.println("<html><body>");
        out.println("<h2>Form Submitted Successfully</h2>");
        out.println("Name: " + name);
        out.println("</body></html>");
    }
}
```

Note: The password is deliberately not displayed in the response.